

Entropy, Cross-Entropy, and KL Divergence

Information theory for generative AI

CSE 534

Goal: connect surprise, uncertainty, and model mismatch

Surprise and information

A rare event is more surprising than a common one.

We want a function $I(x)$ with these properties:

1. rarity gives larger information
2. certainty gives zero information
3. independent events add information
4. small probability changes cause small information changes

These requirements force the logarithm:

$$I(x) = -\log_2 p(x)$$

A likely event gives little surprise; an unlikely one gives a lot.

Self-information

The self-information of an outcome is:

$$I(x) = -\log_2 p(x)$$

Examples:

- ▶ $p(x) = 1/2 \Rightarrow I(x) = 1$ bit
- ▶ $p(x) = 1/4 \Rightarrow I(x) = 2$ bits
- ▶ $p(x) = 1 \Rightarrow I(x) = 0$

This gives a clean measure of surprise in bits.

Entropy: average surprise

Entropy is the expected surprise under a distribution p :

$$H(p) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x)$$

Interpretation:

- ▶ deterministic distribution $\Rightarrow H = 0$
- ▶ more uncertainty $\Rightarrow$ larger entropy
- ▶ uniform distribution maximizes entropy

Maximum entropy

For a finite set of n outcomes, the uniform distribution maximizes entropy:

$$H = - \sum_{i=1}^n \frac{1}{n} \log_2 \left(\frac{1}{n} \right) = \log_2(n)$$

This is the maximum entropy principle: if we know only the set of possible outcomes, a uniform distribution introduces the fewest extra assumptions.

Cross-entropy

Cross-entropy compares the real distribution p with a model distribution q :

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log_2 q(x)$$

Interpretation:

- ▶ if $q = p$, then $H(p, q) = H(p)$
- ▶ if the model is worse, cross-entropy is larger

This is the training objective for language models: minimize surprise on real text.

KL divergence

KL divergence tells us how much extra surprise comes from using q instead of p :

$$D_{KL}(p\|q) = \sum_x p(x) \log_2 \frac{p(x)}{q(x)}$$

A useful identity is:

$$D_{KL}(p\|q) = H(p, q) - H(p)$$

KL divergence is always nonnegative and is zero only when $p = q$.

Why direction matters

KL divergence is not symmetric:

$$D_{KL}(p||q) \neq D_{KL}(q||p)$$

This matters because each direction penalizes different mistakes:

- ▶ $D_{KL}(p||q)$ emphasizes missed real modes in p
- ▶ $D_{KL}(q||p)$ emphasizes mass placed where p has none

In generative AI evaluation, we often choose the direction that matches the failure mode we care about.

Generative AI connection

Training: language models minimize cross-entropy on text data.

For one-hot targets, the loss is just:

$$-\log q(\text{correct token})$$

Evaluation: lower cross-entropy means better predictive fit.

Fine-tuning: KL penalties keep models from drifting too far from the base model.

Takeaway

Self-information measures surprise of one event.

Entropy measures average surprise of a whole distribution.

Cross-entropy measures how surprised a model is by real data.

KL divergence measures the mismatch between two distributions.

These ideas are at the heart of model training, evaluation, and uncertainty in generative AI.